WOLSEY HALL OXFORD
THE HOMESCHOOLING COLLEGE
EST. 1894

SUBJECT	IGCSE Maths Extended
MODULE	One
ASSIGNMENT	One
STUDENT NAME	

ASSIGNMENT BRIEF:

- Answer **all** questions.
- The questions have been written in a similar form to those you will encounter in the IGCSE examination. The questions will require you to demonstrate either some or all of the **two assessment objectives.** These include **AO1** [Knowledge and understanding of mathematical techniques] and **AO2** [Analyse, interpret and communicate mathematically].
- All the resources referred to in the questions are contained within the paper.
- You **must not** use a calculator, unless the question specifies otherwise.
- If the degree of accuracy is not specified in the question, and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place. **Please note that rounding, both premature or in final answers, to 1 or 2 significant figures is penalised in the examination.**
- The number of marks is given in brackets at the end of each question or part-question.
- Your workings **must** be shown with each answer to show your understanding and in order to receive full marks.
- The total number of marks for this paper is **62.** Your mark will be converted to give a percentage.
- Recommended time: **75 minutes.** If you need more time, draw a line across your page and label it ‘time limit reached’, then complete the assignment. This helps your tutor to monitor your progress in time management – an essential exam skill that you will develop over the course.

GENERAL INSTRUCTIONS:

- Make sure you write your full name clearly in the box at the top of this page.
- Please **write** your answers **by hand** in **black ink** on **the question paper**. Drawings which serve to illustrate an answer should be in pencil. Please leave space between your answers for your tutor’s comments.
- You are not expected to complete your assignments without referring to your notes or coursebook. However, we encourage you to rely less on these as you progress through the course. By your final assignment, you should be able to work without reference to any additional notes.
- Please see the Canvas Guide for help with uploading your assignment.

IGCSE Maths Extended Assignment 01

MODULE ONE

This assignment is NON-CALCULATOR for every question. You must show full working for every answer. Answers not supported by working will gain no marks, even if correct.

1. Find:
    a. The lowest common multiple of 3, 30 and 45.

(2 mark)

    b. The highest common factor of 54, 126 and 270.

(2 mark)
(Total 4 marks)

2. A red light flashes every 6 seconds. A yellow light flashes every 4 seconds. They both flash at the same time. After how many seconds will they next both flash at the same time?

.........................seconds

(2 marks)

3. Ian says: ”21 is a prime number”.
Is Ian correct? You must give a reason for your answer.

(1 mark)

4. Write 240 as a product of its prime factors.

(2 marks)

5. Write down the algebraic expression for:
   a. 5 more than x

(1 mark)

   b. y multiplied by z

(1 mark)

   c. 12 less than a

(1 mark)

   d. t divided by 4

(1 mark)

   e. p multiplied by itself

(1 mark)
(Total 5 marks)

6. Evaluate $3x - 2$ when
   a. $x = 0$

(1 mark)

   b. $x = -2$

(1 mark)
(Total 2 marks)

7. Evaluate $\frac{2}{x}$ when
   a. $x = 4$

(1 mark)

   b. $x = -6$

(1 mark)

   c. $x = 0.5$

(1 mark)
(Total 3 marks)

8. Evaluate $x^2 + 2x - 1$ when
   a. $x = 0$

(1 mark)

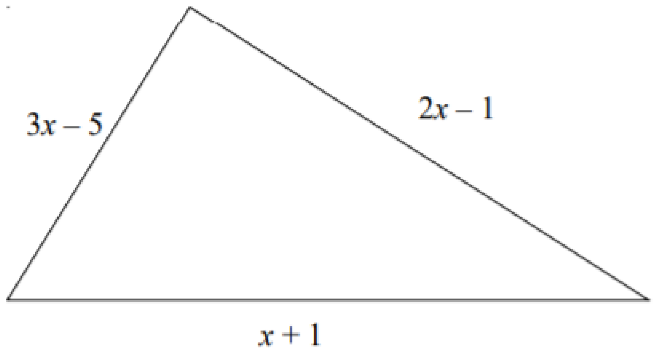
   b. $x = -2$

(1 mark)

   c. $x = 2$

(1 mark)
(Total 3 marks)

9. The lengths, in cm, of the sides of a triangle are $3x - 5, x + 1 \; and \; 2x - 1$.
   Write down an expression, in terms of x, for the perimeter of the triangle. Give your answer in its simplest form.



(2 marks)

10.
   a. Simplify: $6f \; - \; f =$

(1 mark)

   b. Simplify: $7x^2 \; - \; 3x \; + \; 3x^2 \; + \; 6x =$

(2 marks)
(Total 3 marks)

11. Simplify fully: $2(4h + 21) - 5(4h - 1) - 5(h + 1) =$

(3 marks)

12. Simplify fully: $3\left(\frac{2n+7+1}{2}\right) - 5 =$

(2 marks)

13. Find in its simplest form the mean of:

$n + 5,\qquad 6 - 2n,\qquad 2n - 1,\qquad 3n - 4,\qquad n + 9$

(Note: $mean = \frac{sum\ of\ the\ terms}{number\ of\ terms}$)

(4 marks)

14. Simplify fully:

$$\frac{12a}{6ab} =$$

(1 mark)

15. Simplify fully:

$$\frac{x^2y}{z} \times \frac{xz^2}{y^2} =$$

(2 marks)

16. Simplify fully:

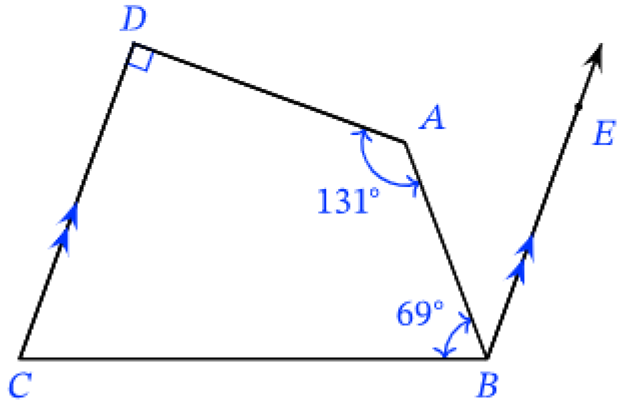
$$\frac{15abc}{5a^2b^2c^2} =$$

(2 marks)

17. What is the size of one interior angle of a regular twenty-sided polygon?

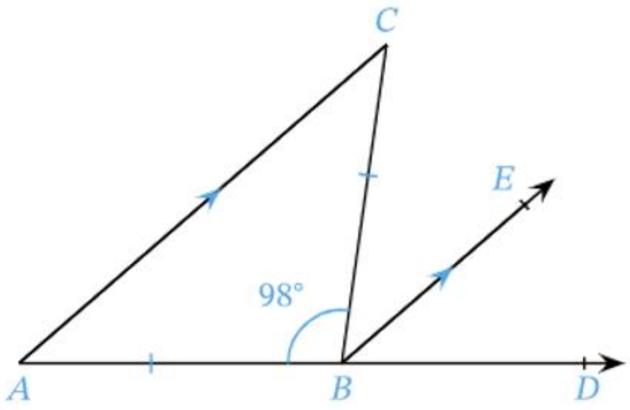
(2 marks)

18. In the figure, CD and BE are parallel. Find the angle ABE.



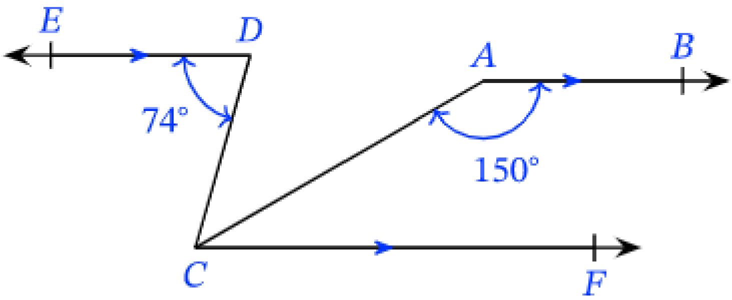
(2 marks)

19. Given that CB = AB, AC and BE are parallel and angle ABC = 98°, determine the measure of the angle EBD.



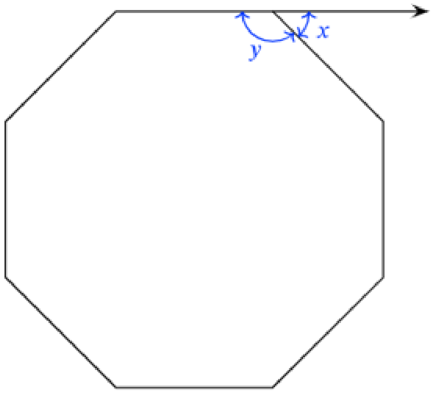
(2 marks)

20. Find angle ACD.



(2 marks)

21. This is a regular polygon. Find x and y. Show full working.



(2 marks)

22. Which polygon has 7 sides and 7 angles?

(1 mark)

23. Write an expression for
    Double the square of x, which is then divided by 9 less than the cube of y.

(2 marks)

24. Explain what is primary and what is secondary data.

(2 marks)

25. How is continuous data normally collected?

(1 mark)

26. Which one of the following is discrete data?
   A Sam is 160cm tall.
   B Sam weighs 60kg.
   C Sam has two brothers and one sister.
   D Sam ran 100m in 10.2 seconds.

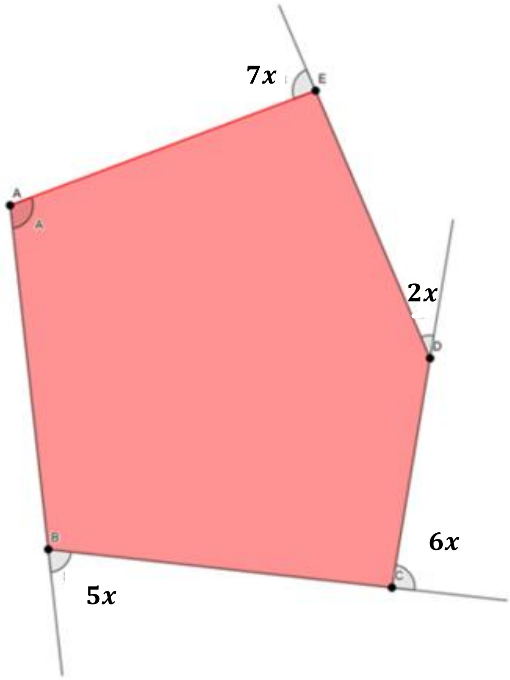
(1 mark)

27. Which one of the following is quantitative data?
   A The dog is black and white.
   B The dog has two ears.
   C The dog has a long tail.
   D The dog is tired.

(1 mark)

28. The irregular pentagon ABCDE has **exterior angles** $5x$, $6x$, $2x$ and $7x$.

    Find an expression for **interior angle A** in terms of x.



(3 marks)

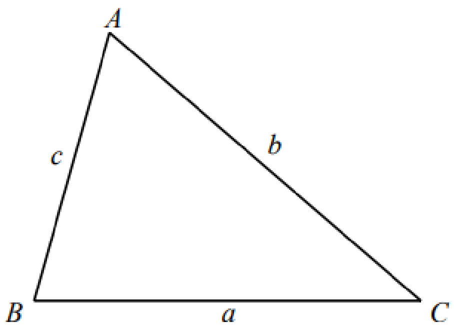
TOTAL FOR ASSIGNMENT 62 MARKS

List of formulas – Extended (Paper 2 and Paper 4)

This list of formulas will be included on page 2 of Paper 2 and Paper 4.

Area, A, of triangle, base b, height h.	$A = \frac{1}{2}bh$
Area, A, of circle of radius r.	$A = \pi r^2$
Circumference, C, of circle of radius r.	$C = 2\pi r$
Curved surface area, A, of cylinder of radius r, height h.	$A = 2\pi rh$
Curved surface area, A, of cone of radius r, sloping edge l.	$A = \pi rl$
Surface area, A, of sphere of radius r.	$A = 4\pi r^2$
Volume, V, of prism, cross-sectional area A, length l.	$V = Al$
Volume, V, of pyramid, base area A, height h.	$V = \frac{1}{3}Ah$
Volume, V, of cylinder of radius r, height h.	$V = \pi r^2 h$
Volume, V, of cone of radius r, height h.	$V = \frac{1}{3}\pi r^2 h$
Volume, V, of sphere of radius r.	$V = \frac{4}{3}\pi r^3$
For the equation $ax^2 + bx + c = 0$, where $a \neq 0$	$x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

For the triangle shown,



$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc\cos A$$

$$\text{Area} = \frac{1}{2}ab\sin C$$